Gopikrishnan D

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ECE Student — AI & Intelligent Systems Developer

Summary

Electronics and Communication Engineering student with a Computer Science minor focused on AI-powered intelligent systems, backend development, and IoT applications. Built real-time systems using Python, FastAPI, OpenCV, Firebase, and embedded platforms. Interested in computer vision, intelligent automation, and scalable software systems.

Education

Rajagiri School of Engineering and Technology — Kakkanad, Kerala
B.Tech in Electronics and Communication Engineering, CS Minor — *Sep. 2023 – May 2027*

Experience

Embedded Systems & IoT Intern — May 2025 – July 2025
NeST Digital — *Kakkanad, Kochi*

- Developed a real-time biometric attendance system using ESP32 and fingerprint sensors.
- Integrated Firebase for cloud-based data management and real-time synchronization.
- Designed and tested complete embedded system including hardware, software, and debugging.
- Performed circuit integration, system validation, and communication protocol testing.

Projects

Smart Attendance System | *Python, FastAPI, PostgreSQL, Docker* | GitHub — Apr. 2026 – Present

- Developed a REST API using FastAPI with JWT-based authentication and role-based access control.
- Designed a relational PostgreSQL database schema covering users, students, and attendance records.
- Implemented duplicate attendance prevention logic for data integrity.
- Containerized and deployed the application using Docker on Render with a live attendance dashboard.

Smart Face Attendance System | *Python, OpenCV, FaceNet, Firebase* | GitHub — Jan. 2026 – Present

- Built a real-time face recognition attendance system using OpenCV and FaceNet.
- Developed a face recognition pipeline using OpenCV for automated student attendance tracking.
- Worked on basic experimentation with FPGA-based deployment workflows for embedded AI applications.
- Built a real-time attendance pipeline integrating camera input, face recognition, and cloud storage.

Smart Biometric Attendance System | *ESP32, Firebase, Embedded C* | GitHub — May 2025 – Jun. 2025

- Built an IoT-based attendance system using ESP32 and a fingerprint sensor.
- Integrated Firebase for real-time cloud synchronization of attendance and fee validation data.
- Built hardware interface with LCD, keypad, LEDs, and buzzer for user interaction.
- Implemented multi-mode authentication and automated attendance logging logic.

Technical Skills

Languages: Python, C, Embedded C, Java, SQL
Backend: FastAPI, REST APIs, PostgreSQL, JWT Authentication, Docker
Databases: PostgreSQL, Firebase Realtime Database
AI / Computer Vision: OpenCV, FaceNet
Embedded & IoT: ESP32, ESP8266, FPGA (PYNQ-Z2), Firebase
Tools: Git, Linux, VS Code

Achievements

Secured high ranks in NTA B.Arch and B.Planning entrance examinations, demonstrating strong spatial reasoning and aptitude skills.

Core Interests

Computer Vision · Intelligent Systems · Backend Development · IoT Automation